Ankita Patel

Research Software Engineer / Software & Data Engineer – Python, ML, HPC

Leicester, UK | 073798 47533 | email.ankitapatel@gmail.com | LinkedIn | GitHub

Professional Summary:

Research Software Engineer and Software/Data Engineer specializing in building robust, reproducible data products and platforms for satellite imagery and sensor-derived datasets. Strong Python background with hands-on experience developing and deploying RESTful APIs, designing scalable data pipelines, and delivering maintainable code with testing, documentation, and CI/CD. Comfortable collaborating with interdisciplinary researchers and external partners, translating requirements into reliable technical solutions, and supporting users through clear communication and training. Research focus: Digital Twins, low-latency data pipelines, and secure service-to-service data exchange.

Technical Skills:

Programming: Python (primary), C#, SQL, JavaScript, HTML/CSS, XML, JSON, Bootstrap
Python (Data/ML): NumPy, Pandas, SciPy, Xarray, scikit-learn, XGBoost, PyTorch/TensorFlow; FAISS (kNN acceleration)
Backend / APIs: Django, FastAPI, Flask; REST APIs
Geospatial / EO: NetCDF/HDF5/native binary formats; GDAL; visualization with Matplotlib/Cartopy
HPC/Performance: JASMIN/ALICE; SLURM (job scheduling, batch submission, resource allocation); profiling; vectorization; chunked processing; parallel I/O; memory control; CMake;
Databases: MySQL, SQL Server, SQLite, MongoDB; Time-series (QuestDB); Vector DB (ChromaDB)
DevOps / Reproducibility: Git/GitHub, GitHub Actions (CI), Docker; testing (pytest); documentation
Platforms / Tools: Linux/Windows; Jupyter, VS Code, PyCharm; Raspberry Pi (2/3/5), Jetson Orin Nano; AWS (S3/EC2/IAM concepts); Grafana; Siemens NX/Teamcenter; MS Office 365
Additional: IDL(basic), Fortran (basic); Excel/VBA; R; Bash Scripting

Professional Experience:

1. **University of Leicester (May 2023 to Present) as Software and Data Engineer**.
   - Contributed to research software across three UK Space Agency–funded programmes (REALM, £680K; CO2SAT, £1.4M; FDSPP ISS Mission), plus administered ~£1M AWS credits for the NCEO EO Data Hub project.
   - **EO / HPC data Pipeline (In-House Research Project)**
     - Refactored legacy EO processing from EUMETSAT Metop-C IDL to Python, cutting wall-clock by ~80% and reducing maintenance risk via type-safe modules, tests, and clearer interfaces. Data format was native binary file with each file size ~1-2 GB and pipeline processes monthly data of 1-2 TB at once.
     - Built a Python-based NASA/NOAA CrIS data pipeline to ingest L1B, L2-CLIMCAPS, and VIIRS products, apply QC and clear-sky filtering, and generate geometry-binned covariance outputs using memory-efficient HDF5 workflows.
     - Designed a configurable Python tool to visualize global Land Surface Temperature (LST) from ESA CCI NetCDF, using Xarray, Dask-based chunked processing, SciPy and Cartopy on JASMIN HPC, to handle large files efficiently and generate high-resolution maps and rotating globe frames.
   - **Digital twin (ESAT-1C) pipeline (In-House Research Project)**
     - Engineered an end-to-end Digital Twin for ESAT-1C satellite featuring automated streaming telemetry real-time timeseries data pipeline (QuestDB ingestion), coupled NX thermal-vacuum solver with ML-based (XGBoost) predictive analytics, and closed-loop feedback system that continuously refines thermal models;
     - Implemented batch processing (160-record batches), comprehensive validation framework, and robust error recovery to ensure reliable 24/7 autonomous operation with minimal intervention.
   - **Synthetic data generation for ML (REALM)**
     - Architected and deployed a high-throughput synthetic satellite imagery pipeline that ingests 40+ GB of GRASP climatology NetCDF data to generate annotated training datasets;
     - Engineered spatially-coherent multi-band scene synthesis with multi-threaded pixel database matching, realistic smoke plume overlays with COCO-format annotations, and adaptive variant blending using bilinear interpolation—enabling 5000+ diverse scene generation while maintaining sub-512MB memory footprint through intelligent caching and resource profiling
   - **Hyperspectral imaging system (CO2SAT)**
     - Developed a Python-based acquisition and processing pipeline for Linescan hyperspectral cameras (Ximea CMV2K-LS150), capturing full 3D data cubes (1088×2048 pixels × 512 spectral bands, 470-900nm).
     - Integrated motorized stages for automated scanning and implemented robust dark/white frame calibration with spectral correction using XML sensor calibration files. Tools for band-specific image display (including O_2 absorption

features at 760nm), false-color composites from user-selected wavelengths, and rapid slice inspection for 3D spatial-spectral exploration.

- **Concurrent Design Facility** (**CDF**) — Comet Budget Tool
  - Developed a configurable spacecraft budget engine (mass/power/data-rate) for Space Park Leicester's Concurrent Design Facility, automating equipment-tree scanning from Excel defined names, dynamic state discovery, k-of-n redundancy modelling, duty-cycle weighting, and margin propagation — enabling multi-disciplinary engineering teams to generate traceable budget tables in a single click.
  - Architected the tool as 8 modular VBA components with structured logging, comprehensive input validation, and conditional-format heatmap visualization, supporting reliable 24/7 use during concurrent design sessions.
- **Embedded Research Software** (**FDSPP-ISS**)
  - Assisted in developing Python-based flight software for a multi-node Raspberry Pi experiment deployed on the ISS via NanoRacks External Platform (NREP).
  - Implemented USB serial communication interface between a Linux host and Raspberry Pi, enabling command and telemetry exchange with the NREP.
  - Validated SPI-based pressure and temperature sensor readouts and LED illumination control (graduated intensity cycling) on experimental hardware, with automated CSV data logging.
- **Research software practices & collaboration**
  - Authored technical documentation, user how-tos, and training materials; collaborated with interdisciplinary researchers and external partners (NCEO, SPL).
  - Established basic CI workflows (GitHub Actions), reproducible builds, structured logging, and secure coding hygiene (AWS IAM awareness).

2. Kartum Infotech (March 2021 – Jan 2022) as Software Developer Intern (Full-time)

- Responsible for assisting the team leader, completing projects, and managing your schedule to ensure deadlines are met. Assist to produce a clean and well-designed self-documenting Code in web applications using **MySQL, jQuery, JavaScript, AJAX, HTML5, CSS3** and back-end programming with object-oriented programming languages like **Python (Django, Flask)**, PHP. Developing backend portals with an optimized database. Troubleshooting application and code issues. Review existing code for coding standards compliance.
- Write technical design documents and keeping track of software developments. Developed and maintained code for in-house and client websites primarily using HTML, CSS, Sass, JavaScript, and jQuery.

Self-Study Projects:

CEH dataset discovery | FASTAPI SvelteKit TypeScript: ETL pipeline + semantic search + AI chat interface; built and deployed REST endpoints for multiple users during experimentation; containerized components using Docker. Link
Cat and Dog Voice Recognizer Model (Audio Classification) | Python TensorFlow: Implemented an audio classification workflow (feature extraction, supervised training, evaluation, prediction on new samples). Link
Skin Cancer Detection Model (Image Classification) | Python TensorFlow: Built an end-to-end lesion image classification pipeline (preprocessing/augmentation, model training, evaluation, inference). Link
Yoga Class Planner (MSc Individual Project) — Python, Django, MySQL: Link

Education:

MSc Advanced Computer Science — University of Leicester, UK (Jan 2020 – May 2022)
Merit (69.2%). Key modules: Foundation of Cybersecurity; Financial Services Information System; Generative Development; Internet & Cloud Computing; Agile Cloud Automation; Mobile and Web application; Software Measurement and Quality Assurance;
BEng Computer Engineering — Veer Narmad South Gujarat University, Surat, India *(Aug 2007 – Jun 2011)*
First Class (63%). Key modules: Information Technology & Management; Database management System; Computer Networks; Information Security & Applications; Internet technology and applications; Software Engineering;

Publications:

Vladimirova, T., Samara-Ratna, P., Tambe, R., Afxentiou, V., Banks, L., Blake, O., **Patel, A.**, Lloyd, S., Maskolenko, A., Jones, M., Ross, D., Vande Hey, J., Fuerte, D. Rapid Environmental Data Extraction for Remote Sensing Using Lightweight Deep Learning Algorithms Onboard Spacecraft. International Astronautical Congress (IAC), 2025. (Paper ID: IAC-25-B1.4.5x101782). [Status: Presented (2025)]

Other Work Experience:

1. Clarendon Mews (June 2022 - June 2023) as **Care Assistance** (full-time)

- Assisted residents with personal care, mobility, meals, and daily routines, ensuring dignity, comfort, and independence.

- Supported residents with severe medical conditions, including late-stage dementia, providing compassionate and specialist care.
- Monitored health and wellbeing, keeping accurate records and reporting changes while offering companionship and emotional support.

2. Precious Hope Health and Home Care (April 2020 – May 2022) as **Care Worker** (Part-time)

- Demonstrated strong **problem-solving skills** by addressing client challenges and adapting care plans to meet individual requirements.
- Enhanced **communication and interpersonal skills** through regular interaction with clients, families, and healthcare professionals.

3. Amar Enterprise (Jan 2018 – Dec 2019) as **E-Commerce Business Development Manager** (Full-time)

- Led organizational changes to implement e-business strategies, achieving targeted growth on B2B (Indiamart, Tradeindia, Alibaba) and B2C platforms (Flipkart, Amazon, Snapdeal).
- Built and maintained strong client relationships through effective communication and tailored solutions, driving account development and sales growth.
- Developed problem-solving strategies to address client needs and improve brand engagement through digital media marketing and innovative branding initiatives.

4. Jani Import-Export (Feb 2016 - Dec 2017) as **B2B Export Executive** (Full-time)

- Designed and executed strategies to secure export orders from international clients, contributing to a significant increase in company revenue.
- Collaborated with cross-functional teams to manage domestic and international shipments, ensuring timely delivery and client satisfaction.
- Utilized **analytical and problem-solving skills** to maintain updated price sheets (FAS, FOB, CFR, CIF) and streamline import/export documentation processes.

5. Fashion Plus Lace (Dec 2013 - Jan 2016) as **B2B Business Sales Executive** (Full-time)

- Managed product inventory and online presence on B2B platforms (Tradeindia, Indiamart), ensuring accurate and up-to-date product listings.
- Built long-term client relationships through effective communication and tailored solutions, driving sales growth and customer retention.
- Demonstrated teamwork by collaborating with marketing and design teams to create compelling product visuals and branding strategies.

6. D.V. Das Manpower (July 2011 - Nov 2013) as **Client Relationship Manager** (Full-time)

- Conducted face-to-face meetings with clients and job candidates, building a robust database and generating new business opportunities.
- Developed communication and negotiation skills through client-facing outreach and servicing.